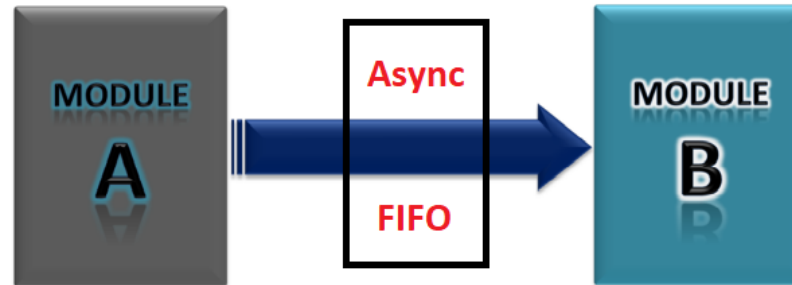


Lab

Example 1 : $f_A > f_B$ with no idle cycles in both write and read.

- Writing frequency (f_A) = 100MHz.
- Reading Frequency (f_B) = 40MHz.
- Burst Length = No. of data items to be transferred = 100.
- There are no idle cycles in both reading and writing which means that, all the items in the burst will be written and read in consecutive clock cycles.



Lab

Solution :-

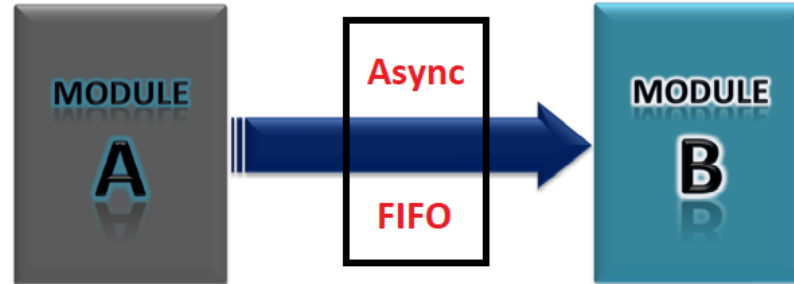
- 1) Time required to **write** one data item = $\frac{1}{100\text{MHz}} = 10 \text{ nsec}$
- 2) Time required to **read** one data item = $\frac{1}{40\text{MHz}} = 25 \text{ nsec}$
- 3) Time required to **write** all the data in the burst = $100 * 10 = 1000 \text{ nsec}$
- 4) Number of data items can be read in a duration of 1000 nsec = $\frac{1000 \text{ nsec}}{25 \text{ nsec}} = 40$
- 5) The remaining no. of bytes to be stored in the FIFO = $100 - 40 = 60$.

The minimum depth of the FIFO should be 60.

Lab

Example 2 : $f_A > f_B$ with idle cycles in both write and read. .

- Writing frequency (f_A)= 100MHz.
- Reading Frequency (f_B) = 40MHz.
- Burst Length = No. of data items to be transferred = 100.
- No. of idle cycles between two successive writes is = 2
- No. of idle cycles between two successive reads is = 2



Lab

Solution :-

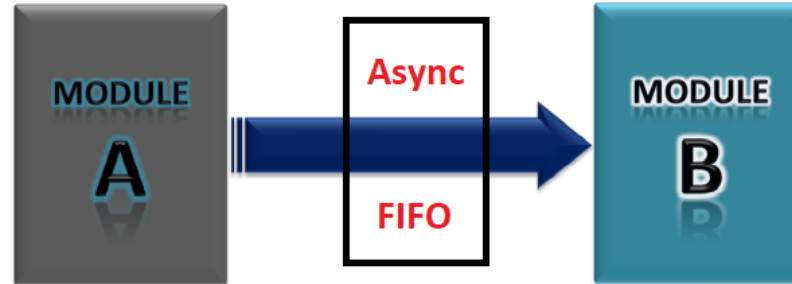
- 1) Time required to **write** one data item = $2 * \frac{1}{100\text{MHz}} = 20 \text{ nsec}$
- 2) Time required to **read** one data item = $2 * \frac{1}{40\text{MHz}} = 50 \text{ nsec}$
- 3) Time required to **write** all the data in the burst = $100 * 20 = 2000 \text{ nsec}$
- 4) Number of data items can be read in a duration of 2000 **nsec** = $\frac{2000 \text{ nsec}}{50 \text{ nsec}} = 40$
- 5) The remaining no. of bytes to be stored in the FIFO = $100 - 40 = 60$.

The minimum depth of the FIFO should be 60.

Lab

Example 3 : $f_A > f_B$ with idle cycles in both write and read. .

- Writing frequency (f_A)= 100MHz.
- Reading Frequency (f_B) = 40MHz.
- Burst Length = No. of data items to be transferred = 100.
- No. of idle cycles between two successive writes is = 2
- No. of idle cycles between two successive reads is = 1



Lab

Solution :-

- 1) Time required to **write** one data item = $3 * \frac{1}{100\text{MHz}} = 30 \text{ nsec}$
- 2) Time required to **read** one data item = $2 * \frac{1}{40\text{MHz}} = 50 \text{ nsec}$
- 3) Time required to **write** all the data in the burst = $100 * 30 = 3000 \text{ nsec}$
- 4) Number of data items can be read in a duration of 2000 nsec = $\frac{3000 \text{ nsec}}{25 \text{ nsec}} = 80$
- 5) The remaining no. of bytes to be stored in the FIFO = $100 - 80 = 20$.

The minimum depth of the FIFO should be 40.